

Assignment

1. Write a C program that will take as input a string, and create a “Sandwich String” from that. A Sandwich String is one where the even indices contain the capitalized characters of that string, and the odd indices contain the same characters in reverse order (check sample input-output for detail).

Sample input	Sample output
String	SGTNRIIRNTGS
Sandwich	SHACNIDWW DINCAHS

Note:

- (a) **You must write a function to generate Sandwich Strings.** You can use the following prototype:
`void sandwich_string(char *str, char *sandwich);`
 - (b) **You must use pointer increment and/or decrement inside this function.** Array indexing is not allowed.
2. Write a C program that will take as input an array, and calculate the sum of the elements of the array so that an element in an even index is added, and an element in an odd index is subtracted. **You must write a recursive function to calculate this result.** You can use this prototype:
`int indexed_sum(int array[], int n);`

Sample input	Sample output	Explanation
8 9 4 3 7 1 5 8 2	3	$9 - 4 + 3 - 7 + 1 - 5 + 8 - 2 = 3$